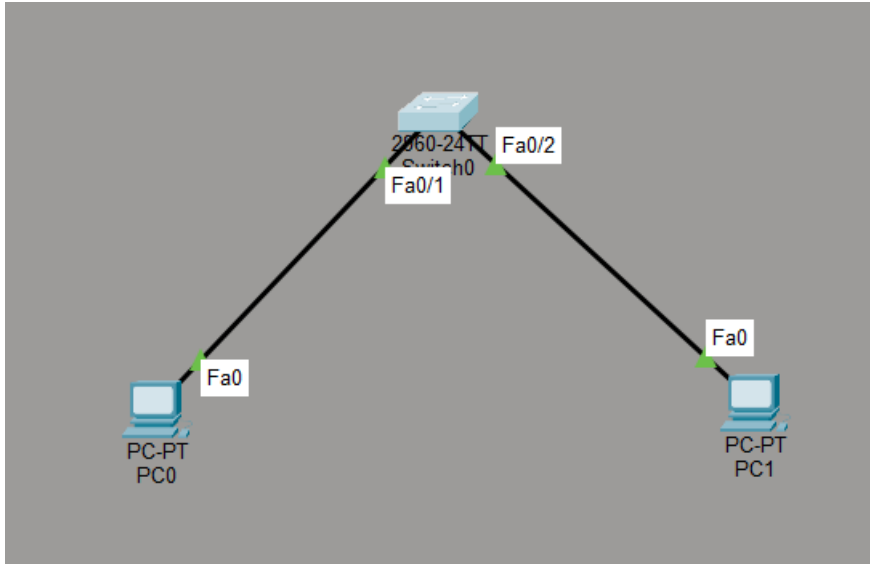


**Q- Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty.**



```
Switch>
Switch>en
Switch#enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
Switch#
```

Initially, the switch does not know which MAC address belongs to which port. When frames pass through the switch, it learns the source MAC address and stores it in the MAC address table along with the incoming port.